

Build from coordinates, not interpretation

This replacement adds actual component holes, mirrored solder views and a numbered endpoint list for every electrical connection.

Fixed construction assumptions

Use **two 24-column x 36-row boards of INDIVIDUAL, ISOLATED pads at 2.54mm pitch**. Coordinates are A-X left-to-right and 1-36 top-to-bottom on the component side. Do not use stripboard or grouped-pad protoboard.

Core Board carries Nano R4, 23LC1024 and ADC filtering. Interface Board carries TL074, power protection, audio and CV circuits. J1 is a direct 2x15 stack connector at H20/I20 through H34/I34 on both boards.

How to read every board page

Mark	Meaning
COMPONENT side	Letters A-X read left to right; red dot marks A1.
SOLDER side	Board is flipped left/right; letters read X-A. Row numbers never change.
Filled dots	Solder endpoints. A crossing line is insulated and does not join other pads.
BARE spine	Solder every named pad on the continuous bus.
J1 key	Remove pin 30 and permanently block socket position I34.

Build order

Stage	Stop point
1	Mark and cut both boards; dry-fit J1 and four standoffs.
2	Place parts and sockets; do not insert ICs or Nano.
3	Complete each solder pass and tick every link ID.
4	Test boards separately, then stack with pin 1 aligned.
5	Run diagnostic firmware before the audio firmware.

Status and hard stop

The coordinate model passes occupied-hole, per-net continuity, J1 agreement and key-position checks. The physical assembly and real-time firmware remain **unbench-tested**. Never use rack power and USB together during first bring-up; open JP1 and separate the boards for USB-only diagnostics.

Complete BOM for the coordinate layouts

Values and reference numbers on this page match the placement drawings. Buy a few spare resistors, capacitors and headers.

Qty	Value	Type	Qty	Hardware	Requirement
10	100n	ceramic or film	1	Arduino Nano R4	Socketed on two 1x15 female headers
1	1u film	ceramic or film	1	23LC1024-I/P	PDIP-8 + socket
1	22n	ceramic or film	1	TL074CP	PDIP-14 + socket
2	1N4148	diode	1 pair	2x15 stack headers	2.54mm; remove/key pin 30
2	1N5817	diode	1	2x10 header + housing	J2 controls
2	BAT85	diode	1	2x4 header + housing	J3 jacks
4	10u	electrolytic	1	2x5 shrouded header	J_PWR Eurorack
1	4.7u	electrolytic	2	24x36 isolated-pad boards	Minimum hole-center area 58.42 x 88.90mm
1	VIN ENABLE	jumper	4	M3 standoffs	10-12mm, plus screws/washers
2	2N3904	transistor			
4	100K	resistor / PTC			
1	100mA PTC	resistor / PTC			
8	10K	resistor / PTC			
9	1K	resistor / PTC			
3	1M	resistor / PTC			
1	250mA PTC	resistor / PTC			
1	39K	resistor / PTC			

- Panel hardware**
- 5 x B10K linear pots; Head 1-3 preferably center-detent.
 - 1 x SPDT ON-OFF-ON quantize toggle.
 - 1 x SPDT ON-ON Loop/One-shot toggle.
 - 2 x normally-open momentary buttons.
 - 1 x 3-lead common-cathode red/green LED.
 - 4 x mono 3.5mm jacks.
 - 10-to-16 Eurorack ribbon cable; red stripe at -12V.
 - 24-28AWG solid insulated hookup wire; crimp contacts for J2/J3.

Part-specific warning: verify the actual 2N3904 E/B/C lead order and electrolytic polarity before insertion. Diode bands and capacitor + marks are printed on placement pages.

Service boundaries and panel harness

J1 replaces inter-board flying wires. J2 and J3 make the panel harness detachable and give every panel lug a named destination.

J1	Net	Role
1	+12P	Core <-> Interface
2	GND	Core <-> Interface
3	+5N	Core <-> Interface
4	GND	Core <-> Interface
5	AUDIO_ADC	Core <-> Interface
6	GND	Core <-> Interface
7	DAC_RAW	Core <-> Interface
8	GND	Core <-> Interface
9	HEAD1	Core <-> Interface
10	GND	Core <-> Interface
11	HEAD2	Core <-> Interface
12	GND	Core <-> Interface
13	HEAD3	Core <-> Interface
14	GND	Core <-> Interface
15	AGE	Core <-> Interface
16	GND	Core <-> Interface
17	MIX	Core <-> Interface
18	GND	Core <-> Interface
19	REC_BTN	Core <-> Interface
20	PLAY_BTN	Core <-> Interface
21	Q_OCT	Core <-> Interface
22	Q_SEMI	Core <-> Interface
23	PLAY_MODE	Core <-> Interface
24	REC_CV	Core <-> Interface
25	PLAY_CV	Core <-> Interface
26	LED_R_IN	Core <-> Interface
27	LED_G_IN	Core <-> Interface
28	+5N	Core <-> Interface
29	GND	Core <-> Interface
30	KEY	Core <-> Interface

J2	Panel net
1	+5N
2	GND
3	HEAD1
4	GND
5	HEAD2
6	GND
7	HEAD3
8	GND
9	AGE
10	GND
11	MIX
12	GND
13	REC_BTN
14	PLAY_BTN
15	Q_OCT
16	Q_SEMI
17	PLAY_MODE
18	LED_R_OUT
19	LED_G_OUT
20	GND

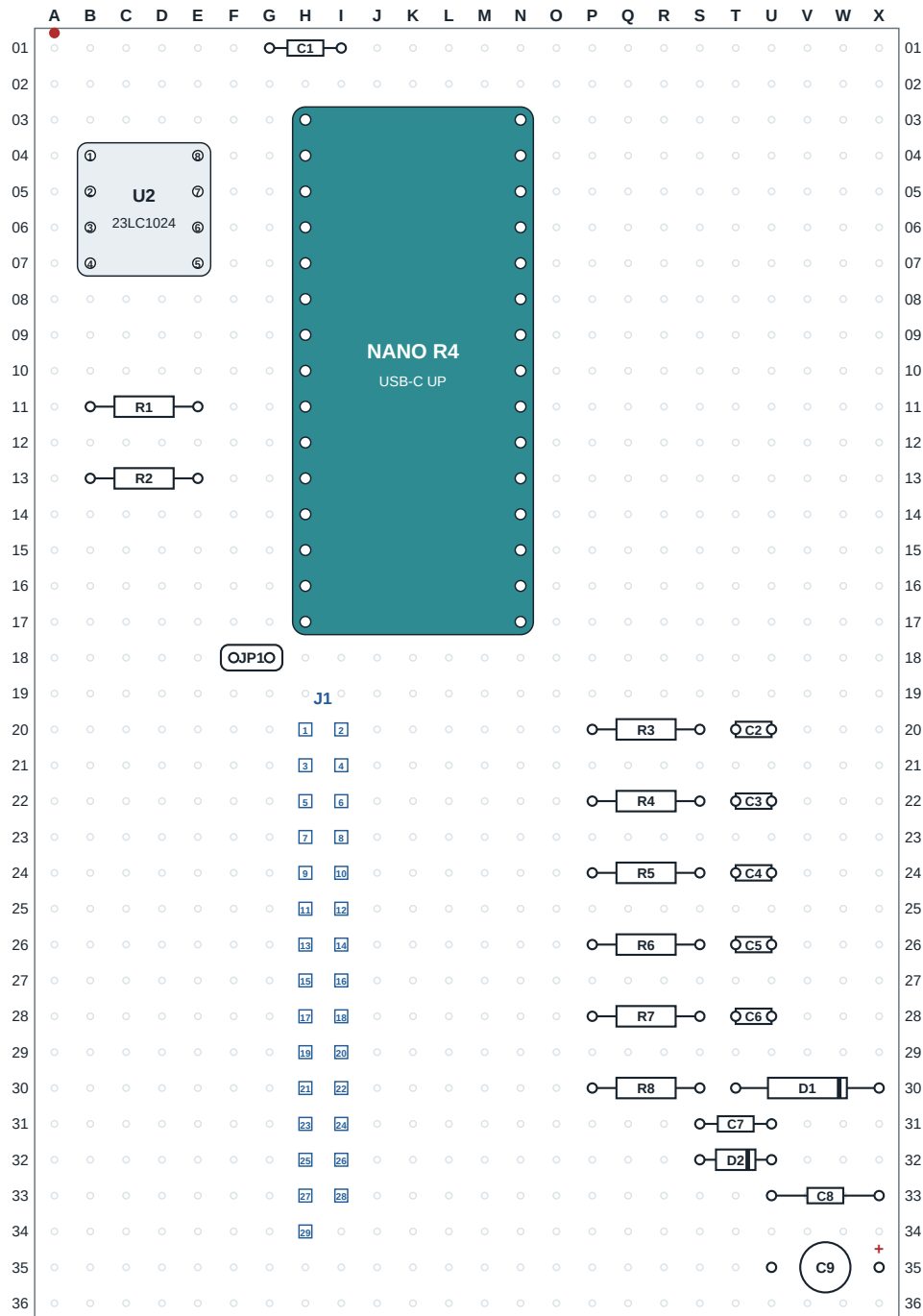
J3	Jack net
1	AUDIO_IN_TIP
2	GND
3	AUDIO_OUT_TIP
4	GND
5	REC_TIP
6	GND
7	PLAY_TIP
8	GND

Orientation

On both boards, J1 pin 1 is H20 and pin 2 is I20. Pin 29 is H34. I34 is the blocked key. Mark H20 red before soldering. J2/J3 pin 1 is the upper-left pad in the component view.

Core Board - component placement

Place every part in the listed holes. USB-C faces row 1. Socket U2 and the Nano; leave JP1 open until power checks pass.



Polarity and orientation

U2 notch faces row 1. D1 band is at X30; D2 band is at S32. C9 positive lead is X35. Nano USB-C faces row 1. J1 pin 1 is H20; I34 stays empty.

Core Board - exact lead and socket coordinates

These tables are authoritative when a label is crowded in the drawing. No two component leads share a hole.

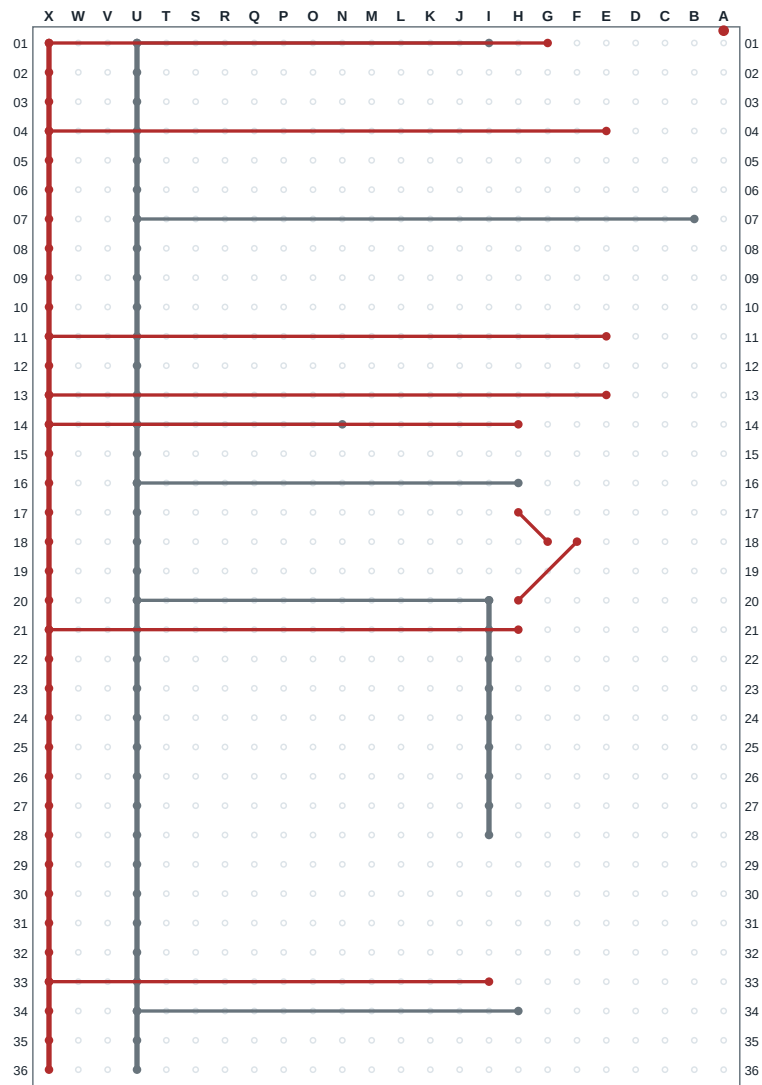
Ref	Value	Lead holes	Nano	Hole	Net
R1	10K	B11 / E11	D13	H3	SCK
R2	10K	B13 / E13	3V3	H4	NC_3V3
C1	100n	G1 / I1	AREF	H5	NC_AREF
JP1	VIN ENABLE	F18 / G18	A0	H6	DAC_RAW
R3	1K	P20 / S20	A1	H7	AUDIO_FILT
C2	100n	T20 / U20	A2	H8	HEAD1_F
R4	1K	P22 / S22	A3	H9	HEAD2_F
C3	100n	T22 / U22	A4	H10	HEAD3_F
R5	1K	P24 / S24	A5	H11	AGE_F
C4	100n	T24 / U24	A6	H12	MIX_F
R6	1K	P26 / S26	A7	H13	NC_A7
C5	100n	T26 / U26	5V	H14	+5N
R7	1K	P28 / S28	BOOT	H15	NC_BOOT
C6	100n	T28 / U28	GND_L	H16	GND
R8	1K	P30 / S30	VIN	H17	VIN
C7	22n	S31 / U31	D12	N3	CIPO
D1	BAT85	T30 / X30	D11	N4	COPI
D2	BAT85	U32 / S32	D10	N5	SRAM_CS
C8	100n	U33 / X33	D9	N6	LED_R_IN
C9	10u	X35 / U35	D8	N7	PLAY_CV
			D7	N8	REC_CV
			D6	N9	PLAY_MODE
			D5	N10	Q_SEMI
			D4	N11	Q_OCT
			D3	N12	PLAY_BTN
			D2	N13	REC_BTN
			GND_R	N14	GND
			RST	N15	NC_RST
			D0	N16	LED_G_IN
			D1	N17	NC_D1

U2 pin	Hole	Name	Net
1	B4	CS	SRAM_CS
2	B5	SO	CIPO
3	B6	SIO2	SIO2
4	B7	VSS	GND
5	E7	SI	COPI
6	E6	SCK	SCK
7	E5	HOLD	HOLD
8	E4	VCC	+5N

J1 coordinates are printed on page 3. The five 1K/100n control filters are R3/C2 through R7/C6. R8/C7 is the audio input filter. BAT85 clamps D1/D2 protect A1.

Core solder pass 1 - ground and power

SOLDER side: columns are mirrored. Build the U-column ground spine and X-column +5N spine before any signal links.



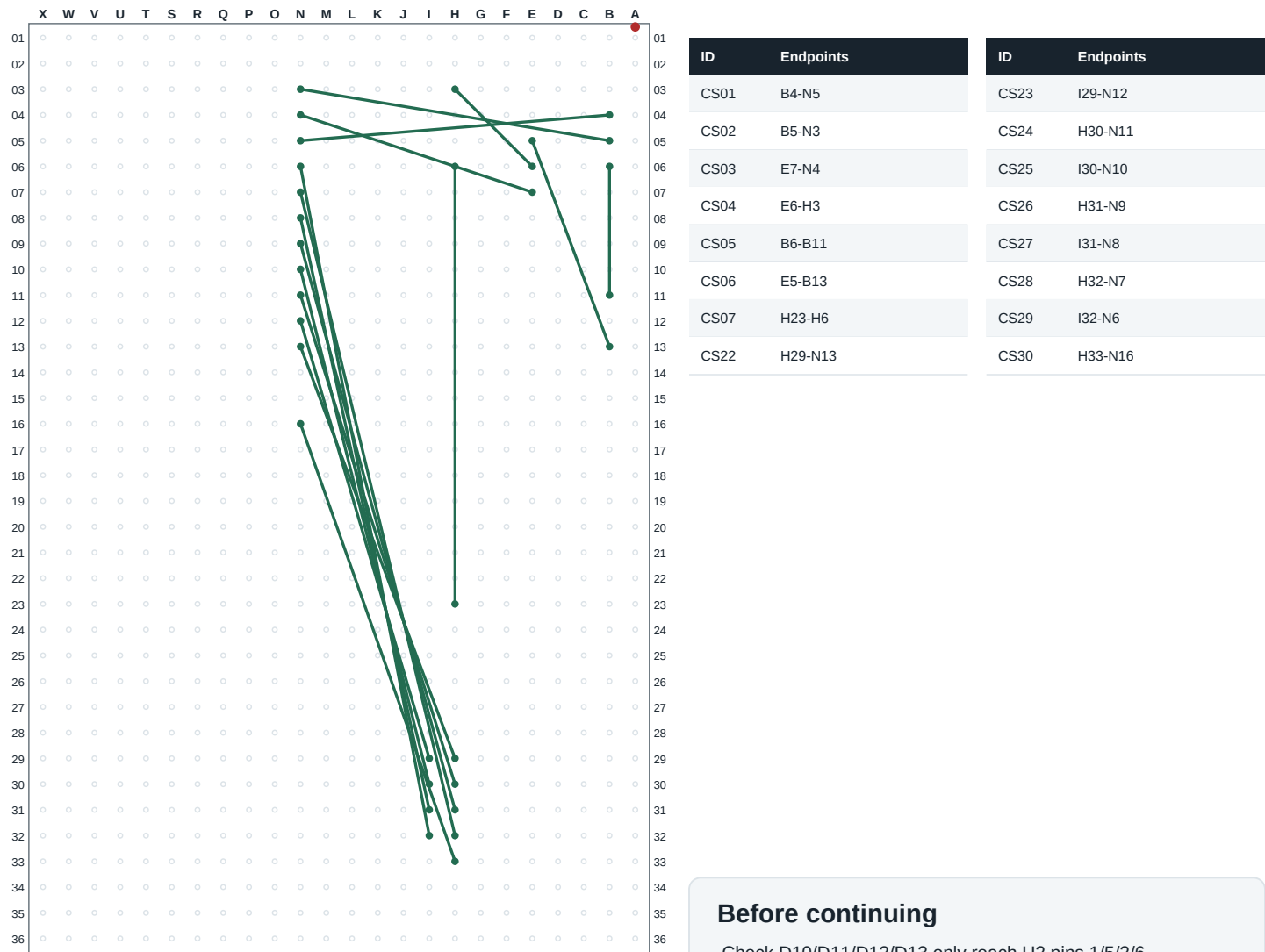
ID	Endpoints	ID	Endpoints
CG01	U1..U36 ALL	CP02	H14-X14
CG02	I20..I28 ALL	CP03	E4-X4
CG03	I20-U20	CP04	E11-X11
CG04	H34-U34	CP05	E13-X13
CG05	H16-U16	CP06	H21-X21
CG06	N14-U14	CP07	I33-X33
CG07	B7-U7	CP07b	G1-X1
CG08	I1-U1	CP08	H20-F18
CP01	X1..X36 ALL	CP09	G18-H17

Checks

CG01 and CP01 are bare spines soldered at every pad.
CG02 is bare only from I20-I28. All other links are insulated.
Verify U-to-X is open before fitting Nano/U2.

Core solder pass 2 - SRAM and digital control

SOLDER side. These links join SPI, buttons, toggles, CV logic and LED control. Crossings are insulated.

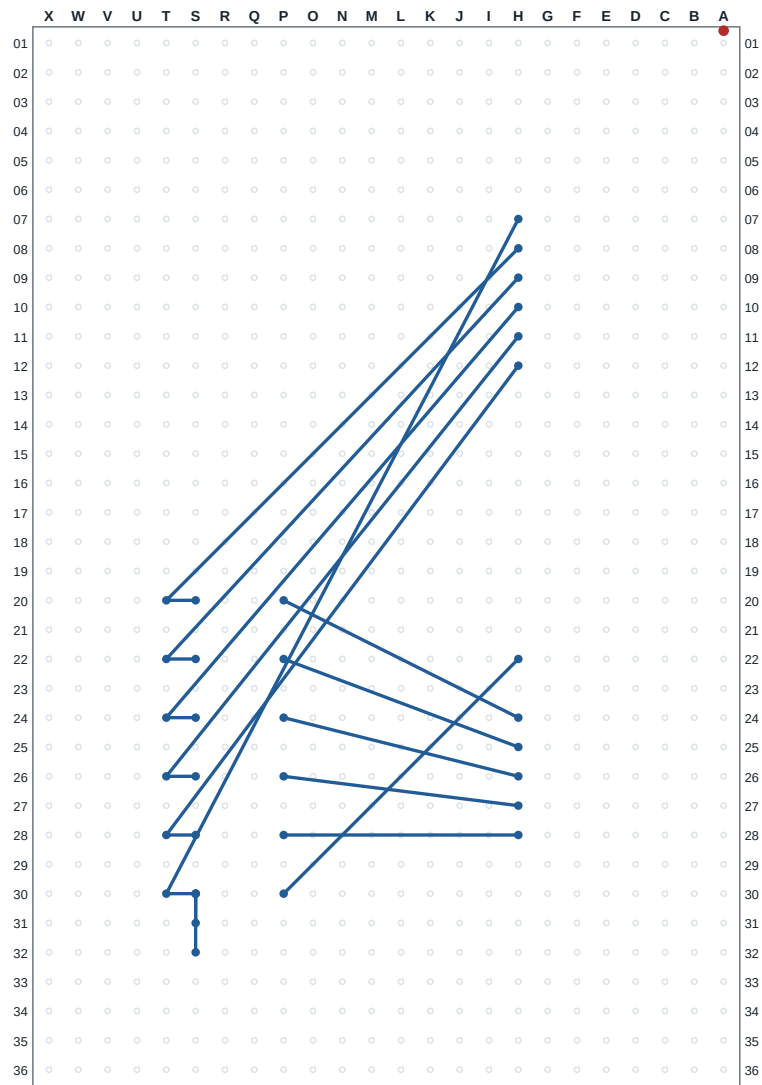


Before continuing

Check D10/D11/D12/D13 only reach U2 pins 1/5/2/6 respectively. U2 pins 3 and 7 reach +5N through R1/R2. No SPI link may touch the ground or +5 spines at a crossing.

Core solder pass 3 - DAC, audio and control ADCs

SOLDER side. Complete CS08-CS21, then meter each J1 analog pin to its Nano destination through the specified resistor.



ID	Endpoints
CS08	H24-P20
CS09	S20-T20-H8
CS10	H25-P22
CS11	S22-T22-H9
CS12	H26-P24
CS13	S24-T24-H10
CS14	H27-P26

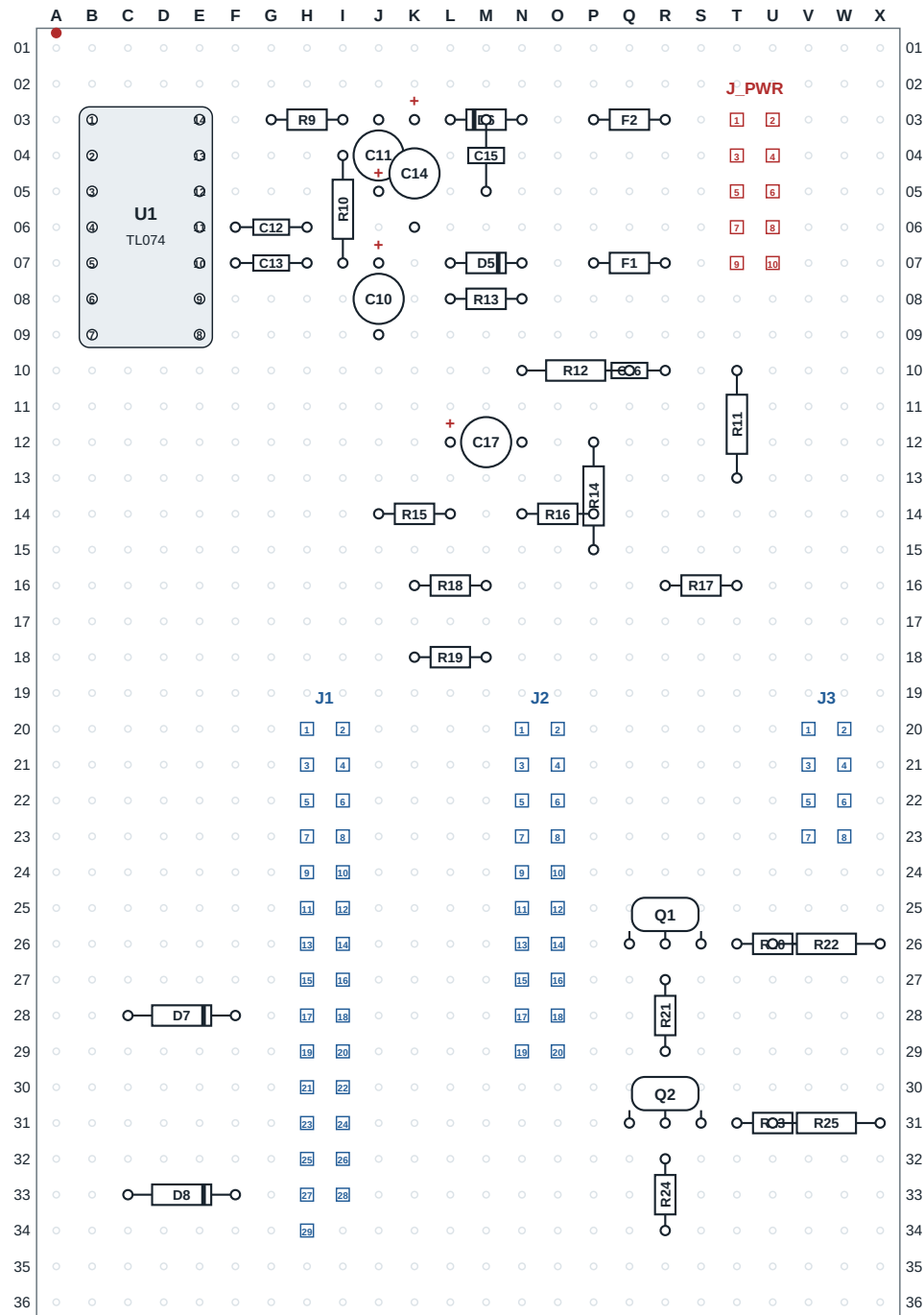
ID	Endpoints
CS15	S26-T26-H11
CS16	H28-P28
CS17	S28-T28-H12
CS18	H22-P30
CS19	S30-T30-H7
CS20	S30-S31
CS21	S30-S32

Expected resistance

J1 pins 9/11/13/15/17 to Nano A2-A6: about 1K.
J1 pin 5 to A1: about 1K.
J1 pin 7 to A0: direct continuity.
D1 cathode: +5N. D2 anode: GND.

Interface Board - component placement

Place U1, protection, audio, CV and header parts exactly as shown. J1 occupies the same coordinates as on the Core Board.



Polarity and orientation

U1 notch faces row 1. D5 band L7; D6 band N3. C10 + at J7; C11 + at J5; C14 + at K3; C17 + at L12. D7 band F28; D8 band F33. Verify Q1/Q2 E-B-C before insertion.

Interface Board - exact lead coordinates, part 1

Power, reference and audio parts. Coordinates refer to the component side even when you are working from a mirrored solder drawing.

Ref	Value	Lead holes
F1	250mA PTC	R7 / P7
D5	1N5817	N7 / L7
F2	100mA PTC	R3 / P3
D6	1N5817	N3 / L3
C10	10u	J7 / J9
C11	10u	J3 / J5
C12	100n	F6 / H6
C13	100n	F7 / H7
R9	10K	G3 / I3
R10	10K	I4 / I7
C14	10u	K3 / K6
C15	100n	M3 / M5
C16	1u film	P10 / R10
R11	1M	T10 / T13
R12	100K	Q10 / N10
R13	39K	L8 / N8
C17	4.7u	L12 / N12

U1 TL074 socket

Pin 1 B3; 2 B4; 3 B5; 4 B6; 5 B7; 6 B8; 7 B9. Opposite row: pin 14 E3; 13 E4; 12 E5; 11 E6; 10 E7; 9 E8; 8 E9. Notch faces row 1.

Interface Board - exact lead coordinates, part 2

Output, LED and CV-trigger parts. Transistor holes are emitter, base, collector only if your actual 2N3904 matches that lead order.

Ref	Value	Lead holes
R14	100K	P12 / P15
R15	10K	J14 / L14
R16	10K	N14 / P14
R17	1K	R16 / T16
R18	1K	K16 / M16
R19	1K	K18 / M18
R20	100K	V26 / T26
R21	1M	R27 / R29
D7	1N4148	C28 / F28
Q1	2N3904	Q26 / R26 / S26
R22	10K	U26 / X26
R23	100K	V31 / T31
R24	1M	R32 / R34
D8	1N4148	C33 / F33
Q2	2N3904	Q31 / R31 / S31
R25	10K	U31 / X31

Headers on the component side

J1: H20/I20 through H34/I34; I34 empty.

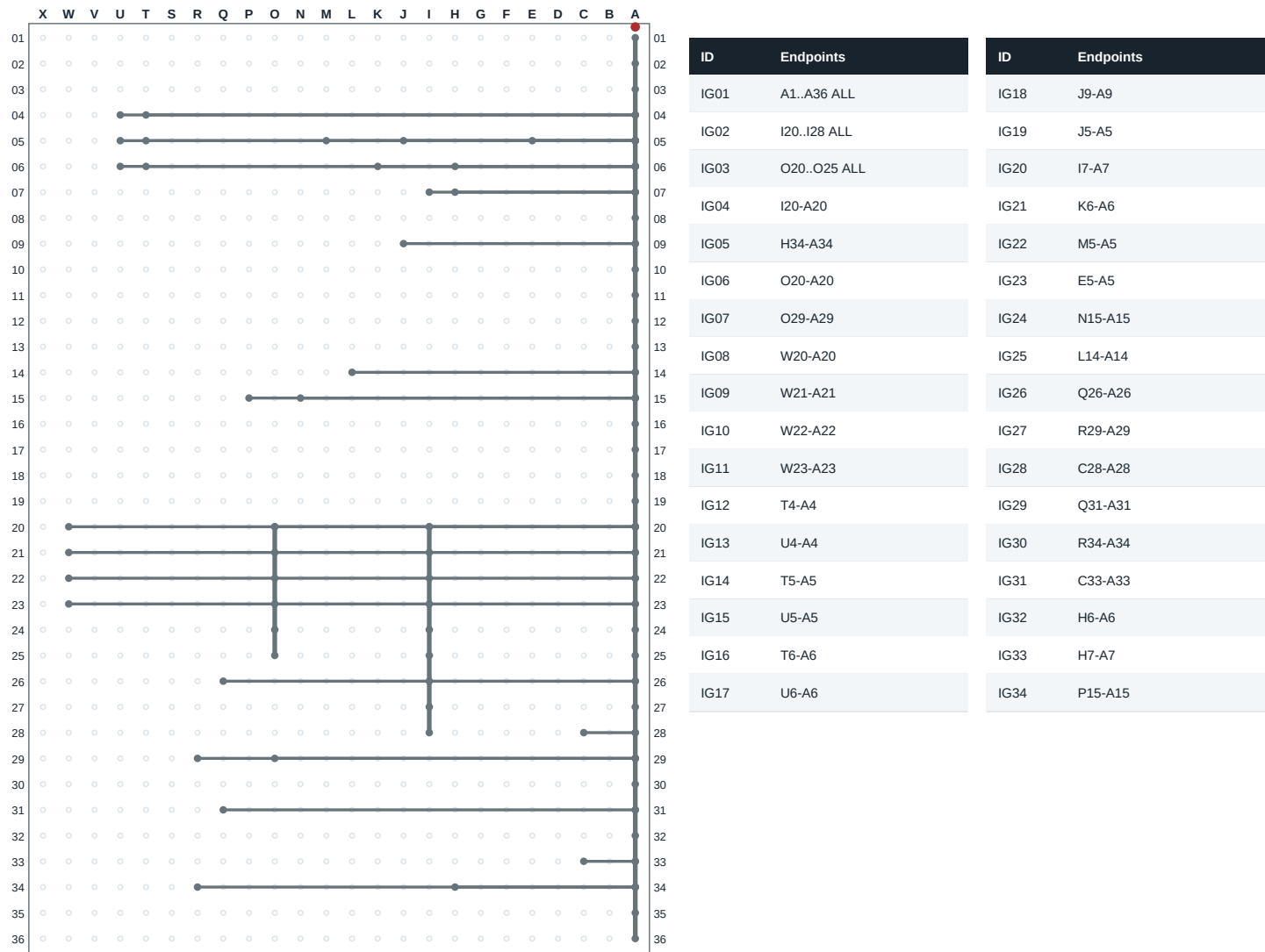
J2: N20/O20 through N29/O29.

J3: V20/W20 through V23/W23.

J_PWR: T3/U3 through T7/U7; keyed notch must force red stripe to pins 1/2 at row 3.

Interface solder pass 1 - ground

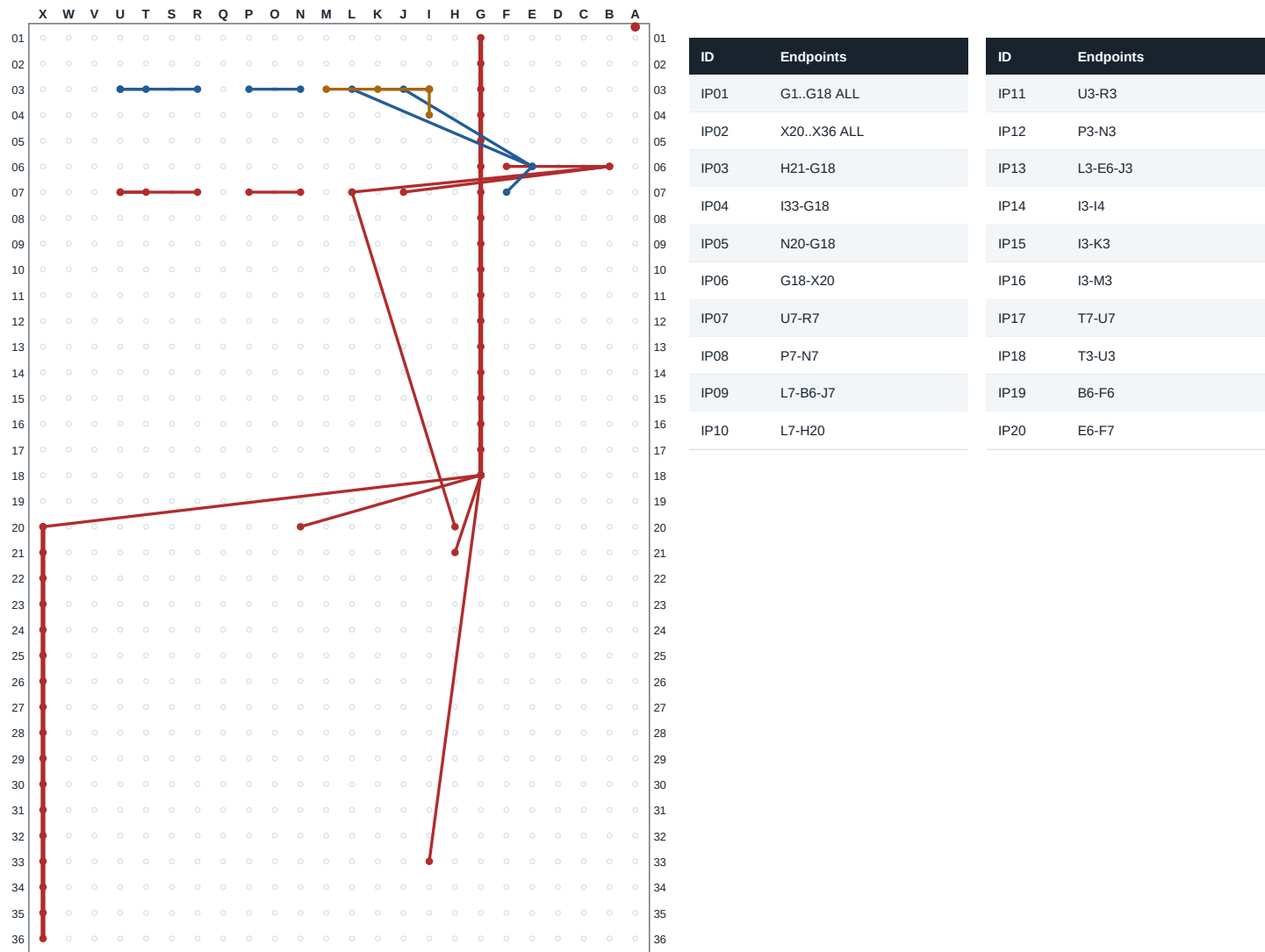
SOLDER side. A1-A36 is the main bare ground spine; J1 and J2 each have a shorter local bare ground spine.



Continuity
Every J_PWR ground pin, every jack sleeve, U1 pin 12, J1 grounds and J2 grounds must reach A1-A36. No rail or signal may reach A.

Interface solder pass 2 - rails and VREF

SOLDER side. Red is +5N/+12; blue is -12; amber is the 2.5V reference divider node.

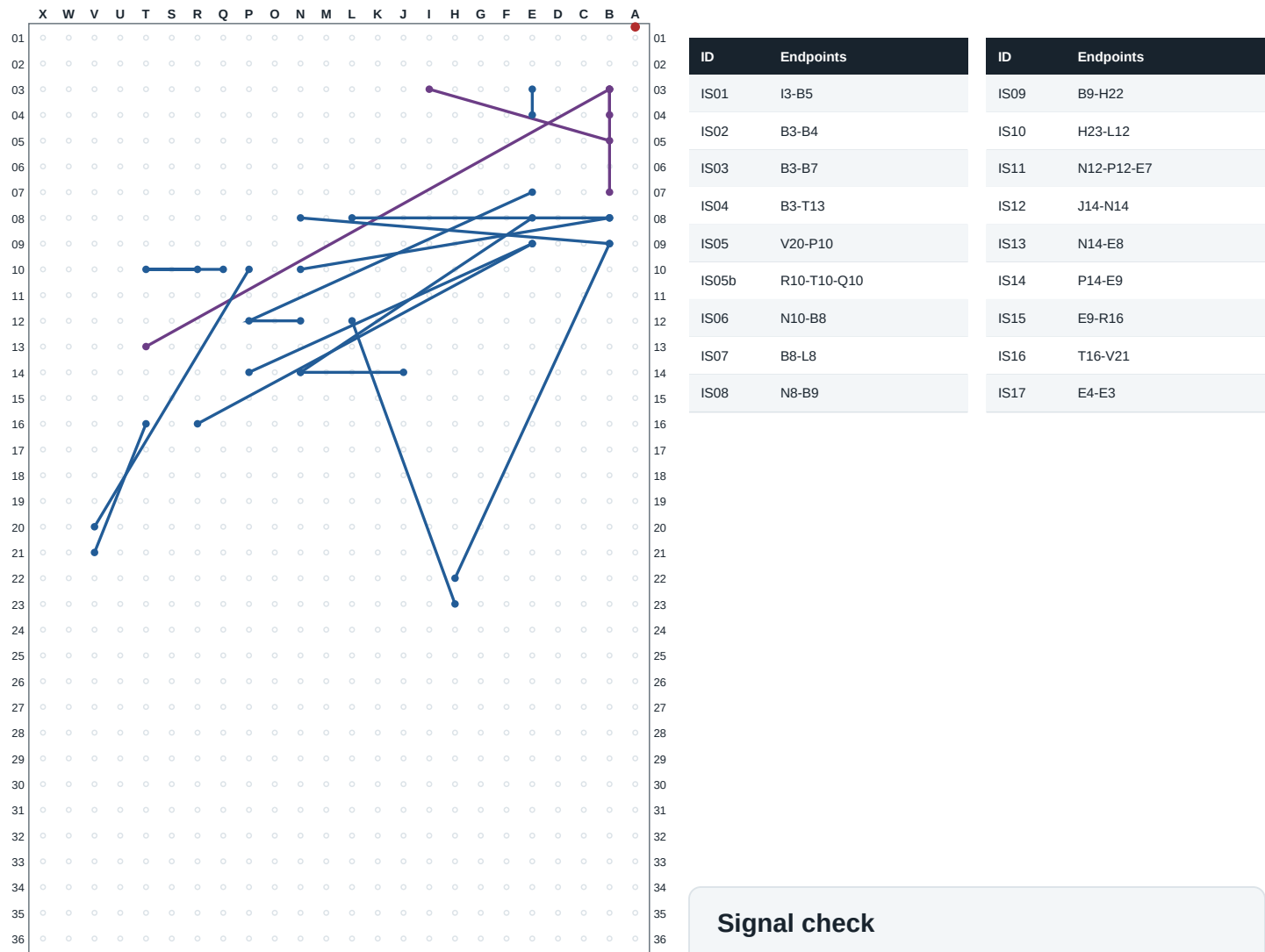


Power before ICs

With J1 disconnected: verify protected +12 at L7/J7/B6, protected -12 at L3/J3/E6, and no short to A/GND. After Nano supplies +5N through J1, VREF_DIV must be near 2.5V.

Interface solder pass 3 - reference and audio

SOLDER side. These are the U1 reference, input attenuation, DAC coupling and output-gain links.

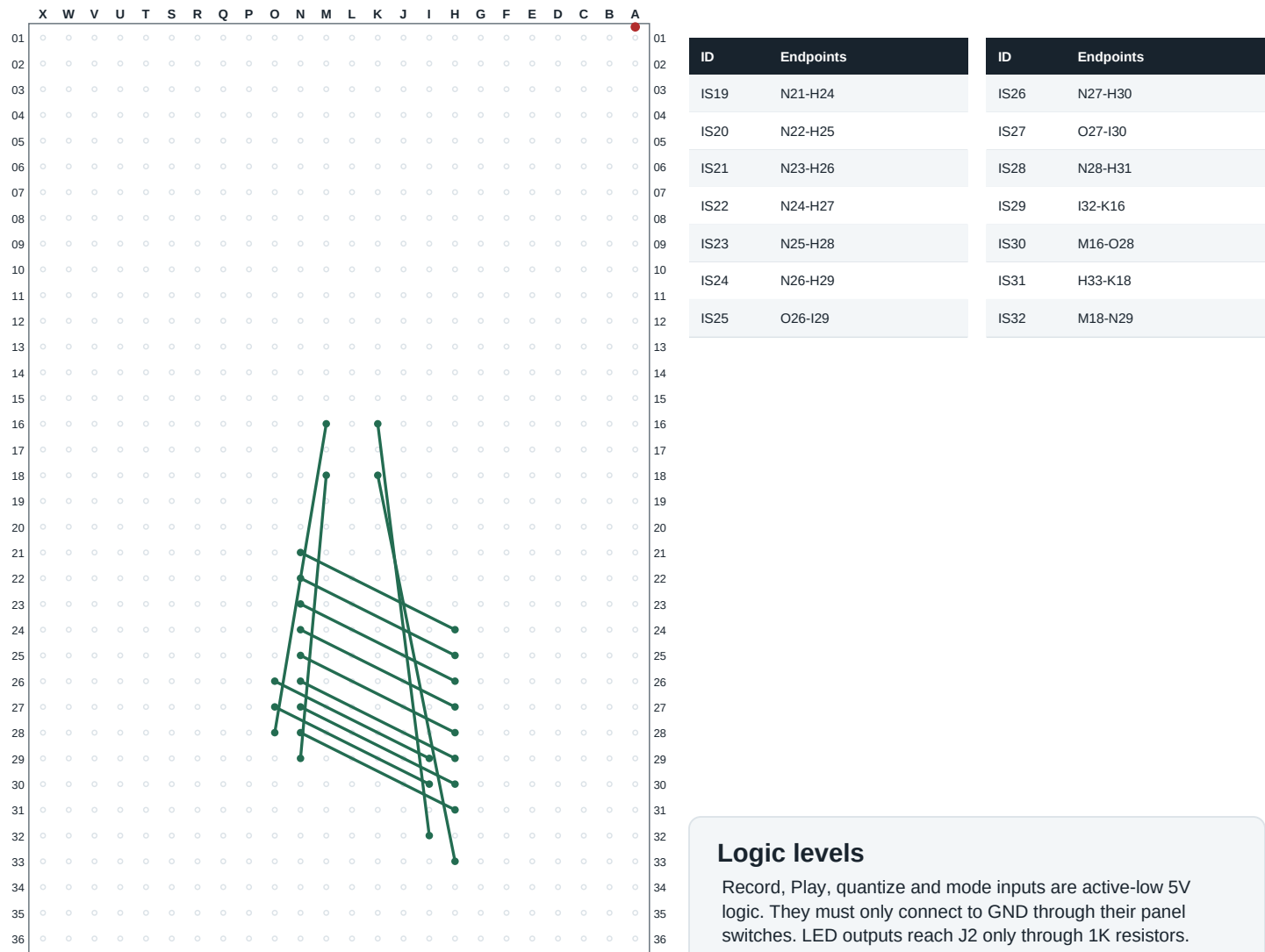


Signal check

No input: U1 pin 7 and J1 pin 5 should rest near 2.5V. J3 audio-output tip should rest near 0V DC after C17. U1 pin 14 is tied to pin 13; pin 12 is grounded.

Interface solder pass 4 - controls and status LED

SOLDER side. J2 signals route to J1; R18/R19 limit the red and green LED currents.



Interface solder pass 5 - Record and Play CV

SOLDER side. Two identical NPN stages convert positive Eurorack pulses to clean active-low Nano inputs and clamp negative CV.



ID	Endpoints	ID	Endpoints
IS33	V22-V26	IS37	V23-V31
IS34	T26-R26-R27	IS38	T31-R31-R32
IS35	F28-R26	IS39	F33-R31
IS36	S26-U26-I31	IS40	S31-U31-H32

Transistor and diode check

Q1/Q2 holes left-to-right are Q-R-S = emitter-base-collector in the component drawing. Confirm your actual 2N3904 before insertion. D7/D8 anodes reach GND; bands face the base nodes. Collector outputs idle HIGH through R22/R25.

Panel hardware - exact lug connections

The panel harness is detachable at J2/J3. Lug names are functional; verify physical pot and jack lugs with a meter before crimping.

Part	Exact connection
HEAD 1	CW J2.1; wiper J2.3; CCW J2.4
HEAD 2	CW J2.1; wiper J2.5; CCW J2.6
HEAD 3	CW J2.1; wiper J2.7; CCW J2.8
AGE	CW J2.1; wiper J2.9; CCW J2.10
MIX	CW J2.1; wiper J2.11; CCW J2.12
RECORD	NO contact J2.13; other contact J2.20 GND
PLAY	NO contact J2.14; other contact J2.20 GND
QUANTIZE	common J2.20 GND; throws J2.15 OCT / J2.16 SEMI
LOOP/SHOT	common J2.20 GND; one throw J2.17; other throw unused
LED	red anode J2.18; green anode J2.19; common cathode J2.20

Jack	Connection
AUDIO IN	tip J3.1; sleeve J3.2
AUDIO OUT	tip J3.3; sleeve J3.4
RECORD CV	tip J3.5; sleeve J3.6
PLAY CV	tip J3.7; sleeve J3.8

Suggested 14HP centers

Millimetres from panel upper-left:

H1 14,20; H2 35.4,20; H3 56.8,20.
Age 22,45; Mix 49,45.
Quant 18,67; Mode 45,67.
Record 18,84; LED 35.4,84; Play 53,84.
Jacks x 9/26/44/61 at y108.

Fit rule

These centers are a layout aid, not a blind drill template. Confirm bushing diameters, body clearance, rail clearance and the actual 14HP panel width before drilling.

Staged power-up and functional checks

Do not insert the Nano, U1 or U2 until the unpowered checks pass. Power down immediately if any voltage or temperature is wrong.

A. Boards separated, unpowered

- ☐ Both I34 key positions are empty/blocked; H20 is marked pin 1 on both boards.
- ☐ No short from +12P, -12P or +5N to GND; +12P and -12P are not shorted together.
- ☐ Core J1 pins 9/11/13/15/17 reach Nano A2-A6 through about 1K.
- ☐ Interface J_PWR pins 1/2 reach only the -12 path; pins 9/10 reach only the +12 path.

B. Interface Board alone

- ☐ Apply current-limited rack power: +12P and -12P have correct polarity after D5/D6.
- ☐ Insert U1 only after rails pass. Confirm VREF output U1 pin 1 is near 2.5V when +5N is present.
- ☐ AUDIO_ADC rests near 2.5V; AUDIO OUT tip rests near 0V DC.

C. Stacked diagnostic

- ☐ Fit JP1, stack J1 squarely, power from rack, and verify Nano VIN and +5N before fitting U2.
- ☐ Diagnostic firmware reports all pots, buttons and toggles correctly; CV inputs idle HIGH and pulse LOW.
- ☐ Fit U2 and pass a walking-byte test across addresses 0x00000-0x1FFFF.

D. Audio firmware

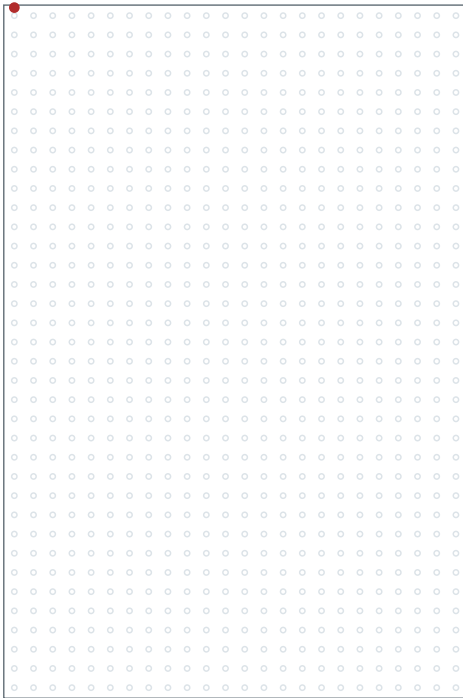
- ☐ Mix fully CCW is dry-only; fully CW is wet-only.
- ☐ All three head pots at detent give stable +1x playback.
- ☐ Record/overdub, Loop/One-shot, manual retrigger and both CV inputs match the firmware contract.

USB rule

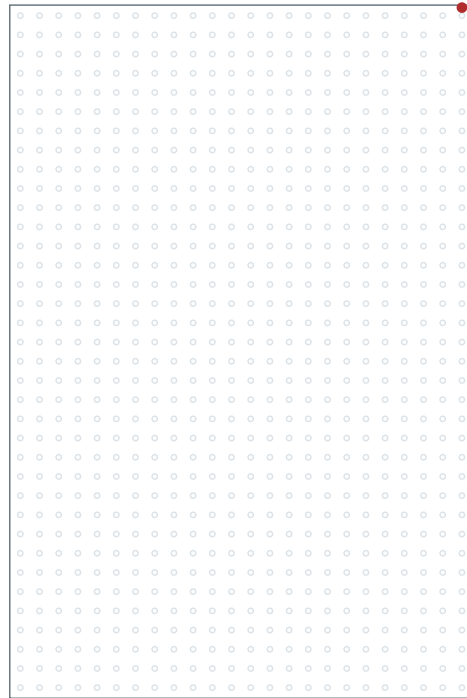
For USB-only flashing: remove the module from the rack, open JP1 and unplug J1. Do not let USB back-power the unpowered TL074 board through signal clamps.

Print scale, board marks and build record

Only the two grids below are actual 2.54mm pitch at 100% printing. All earlier board drawings are enlarged.



COMPONENT



SOLDER

Print at 100% / Actual size. The line below must measure exactly 50mm.

50mm



Build record

Builder: _____ Date: _____ Board rev: _____

+12P: _____V -12P: _____V +5N: _____V VREF: _____V

Firmware file / commit: _____

SRAM: PASS / FAIL Audio: PASS / FAIL Controls/CV: PASS / FAIL

Notes: _____

Primary references

Arduino Nano R4: <https://docs.arduino.cc/hardware/nano-r4>

Microchip 23LC1024 DS20005142: <https://www1.microchip.com/downloads/en/DeviceDoc/20005142C.pdf>

Doefer A-100 power pinout: https://doefer.de/a100_man/a100t_e.htm